

High-Level System Design

MAINNET TARGET
TESTNET VERIFIED

The Daily Walrus is a wallet-gated AI football app where every important action gets a signed Sui proof, Walrus stores the durable payloads, and Walrus Memory gives Gil recall across sessions.

Architecture Canvas



Wallet + SPA

React/Vite on Walrus Sites.
Connects Sui wallet, signs prediction/output actions, then renders proof links for judges.



Gil Agent API

Mastra server with fixture tools, prediction tools, roast generation, memory recall, and output anchoring.



Walrus Memory

Canonical memory namespaces: global fixtures/team facts and per-wallet notebooks.



Walrus Blob

Immutable media and JSON payloads: posters, gallery art, team profiles, output records.



Sui Objects

Owned prediction gates, result settlement, score records, and blob/hash receipts.



Index Cache

Supabase indexes fixtures, leaderboard, proof URLs, and rebuildable UI state.

Wallet Boundary

No anonymous writes. Prediction, roast, notebook, and share outputs start from a connected wallet session.

Memory Boundary

Walrus Memory is the source of truth for Gil's recall. Supabase can be rebuilt from proofs and blobs.

Contract Boundary

Sui stores compact proofs: object IDs, blob IDs, content hashes, gates, scores, and settlement state.

Judge Boundary

Runtime Tracking exposes live network, package, object, memory, blob, and explorer links.

Design rule: store big content on Walrus, store recall facts in Walrus Memory, store public receipts and gates on Sui. The app can be fast without pretending the cache is the source of truth.

Signed Action + Data Flow

SHORT DESIGN
SUBMISSION PDF

Four flows cover the product: prediction, chat/roast output, global memory refresh, and judge verification. Each flow has one clear owner and one clear proof surface.



System Components

- Web SPA: prediction desk, chat, gallery, team profiles, tracking pages.
- Agent API: Mastra tools for fixtures, memory recall, roasts, and output records.
- Sui package: prediction registry, lock/settle rules, score and output proof objects.
- Walrus: blobs for media/payloads, Walrus Memory for static and wallet-scoped recall.

Requirement Fit

- Persistent AI memory: Walrus Memory, split by global and wallet namespace.
- On-chain verifiability: Sui object receipts link to blob IDs and content hashes.
- Prediction integrity: match gates close before kickoff and settle after results.
- Submission proof: Runtime Tracking lists deployed contract, object, memory, and site links.